

ELSAM RAFI SAPUTRA

Bandung, Indonesia | Email: elsamrafisptra@gmail.com | [LinkedIn](#) | [Github](#) | [Portfolio](#) | [Website](#)

Work Experience

SCHNAP – Australia's premier electrical wholesaler and supplies

Kingsgrove, Australia (Remote)

Software Engineer

March 2025 - Now

- Optimized product data pipeline (130K+ products), increasing average product success rate from <45% to 71% and still get bigger every time by combining Breadth-First Search (BFS) and Backtracking algorithms.
- Optimized product ingestion pipeline by implementing asynchronous processing, batch inserts, and data validation, reducing ingestion time from 2 hours to 45 minutes and being faster on the next ingestion.
- Implemented and optimized SEO-driven product frontend, improving SEO score from <70 to >85 and contributing to a 250% increase in paid transaction revenue within six months.
- Developed an adaptive rate limiter for development environment to dynamically throttle requests during high latency or 5xx server errors, reducing production error rates and downtime by 50%.
- Integrated and maintained AI agents for product enrichment, marketing automation, and internal tools through prompt engineering and context optimization using Gemini and OpenAI APIs.
- Optimized frontend performance (e-commerce) by implementing lazy loading and caching.
- Developed product monitoring dashboard with Vue.js and Laravel for development internal tools.
- Built pdf extraction tools for pipeline integration using python, OCR-based extraction, content parsing, and context engineering using Open AI inspired by [Mineru Document Content Extraction](#).
- Stack: Golang (Gin), Python, Vue JS, Playwright, MySQL, AWS, CS Cart.

KAP TNN – Indonesia's registered public accounting firm

South Jakarta, Indonesia (Remote)

Fullstack Developer

September 2024 – August 2025

- Developed internal document management platform ("Datatrail") supporting multi-tenant architecture and automated workflow processing using Next JS, FastAPI and MySQL.
- Optimized frontend performance and SEO implementation, elevating the company's website ranking from beyond #100 to Top 10 (first page) for target keywords, improvements in Core Web Vitals, structured metadata, and Lighthouse score optimization.
- Built multiple automation pipelines with Python, Make.com, Zapier, and n8n, reducing manual workloads and company operational costs by 20%.
- Integrating third-party APIs (OpenAI, Azure, PDF.co, GCP, etc) for data, webhooks, and another tasks.
- Developed custom WordPress plugins using PHP (MVC pattern) for internal features and external APIs.
- Implemented mathematical, accounting, and finance modules leveraging NumPy and SciPy for the features.
- Stack: Next JS, FastAPI, MySQL, Azure, Make.com, Zapier, n8n, Wordpress.

School Tech Indonesia – Digital school management startup

Malang, Indonesia (Remote)

Frontend Developer

August 2024 – December 2024

- Developed "Internpro", a digital school management and internship platform, implementing modular and reusable UI components using Ant Design and React to enhance scalability and performance.
- Integrated Laravel RESTful APIs with frontend components for real-time data synchronization, authentication, and system management features.
- Wrote unit tests with Jest to validate API interactions and improve frontend reliability.
- Stack: React JS, Ant Design, Jest, Laravel, MySQL.

Betr Beta – Software agency

Anson Road, Singapore (Remote)

Fullstack Developer

July 2024 – August 2024

- Optimized backend–frontend data fetching performance using the Render-as-you-fetch pattern and React Suspense, significantly reducing page load times and improving API response efficiency.
- Diagnosed and resolved backend logic bugs in the Release Notes Manager features by implementing end-to-end test cases using Cypress + Cucumber.
- Stack: Next JS, Shadcn/ui, Cypress, Firebase.

Organizational Experience

Telkom University Education Movement – Social & Education

Bandung, Indonesia

Design and Publication Team

October 2021 – August 2022

- Designed social media themes, and Instagram feed templates, and organization programs/events.
- Participated in volunteering programs: “Mengajar: Road to Desa 2022” in Cimaung village, Extracurricular as an English teacher at SDN Cipagalo 3 for two months, and Aksi Peduli Desa FKB Universitas Telkom.

Menebarmanfaat.id – Social & Personal Development

Bandung, Indonesia (Remote)

Design and Motion Video Division

November 2021 – March 2022

- Designed daily feeds, short videos, other organization program designs.
- Participate in every work program activity in the online activities trends during the COVID-19 era.

Projects

StuntFree Stunting Prevention and Detection Apps – Bangkit Academy 2024

- Designed separated main services and ML services in Cloud Run with docker and github actions CI/CD.
- Led mobile developer team by designing and compiling the majority of the app’s UI/UX.
- Provided approximately 25 datasets from scrapping various sources and streamlining data preparation for fine-tuning main ML model training on Vertex AI.
- Built simple models using KNN for classification environment quality case to support main ML models.
- Developed Restful API as main services and gRPC server to load and serve machine learning model and integrate it with main services and Vertex AI.

Growth Momentum (E-Learning Platform) – Final Project (ICADEIS 2025)

- Integrated machine learning models (recommendation system) using Google Collab and dataset from Kaggle (E-Learning) using K-Means Clustering by calculating Novelty and Diversity Metrics in Flask backend.

Teman Anti Judi Online – Microsoft Online Hackathon 2025 (elevAlte with Dicoding)

- Developed AI personalized feature by using Azure AI Personalizer, Azure Open AI, and pre-trained models and dataset from hugging face.
- Developed AI chatbot by integrating Azure Open AI and Azure AI Content Safety.
- Developed negative sentiment media for online gambling by scraping from news media, blog media, and social media by using AI Content Safety and Azure Open AI.

Education & Achievements

Telkom University

Bandung, Indonesia

Bachelor of Informatics (GPA 3.77/4.00) – Cumlaude

September 2021 – January 2025

- Final Project’s title “Evaluation of Diversity and Novelty of Recommendation System on Digital Learning Platform Based on User Usability and Satisfaction” Accepted at International Conference ICADEIS 2025.
- Programming Algorithm & Data Structures Practicum Assistant at Informatics Laboratory.
- Telkom University Endowment Fund Scholarship 2024.

Bangkit Academy 2024 By Google, GoTo, Tokopedia, Traveloka

Jakarta, Indonesia (Remote)

Cloud Computing Cohort (A, 87/100)

September 2024 – January 2025

- Completed the Cloud Computing Path, including courses in cloud infrastructure, frontend and backend development, Google Cloud tools, Google Cloud Skills Boost, Dicoding, and Coursera courses.

- **elevAlte with Dicoding (Microsoft Azure) 2025**

- **AWS Academy 2024 Backend Development Stage 2**

Skills: Typescript, Python, Golang, PostgreSQL, MySQL, Firebase, MongoDB, Next JS, Express JS, FastAPI, gRPC, Flask, Wordpress, Cloud Computing, Problem Solving, Algorithms & Data Structures.